ECE279:BASIC ELECTRICAL AND ELECTRONICS ENGINEERING LABORATORY

L:0 T:0 P:2 Credits:1

Course Outcomes: Through this course students should be able to

CO1 :: assemble Assemble various measuring instruments and electronic components on the breadboard and create circuit connections.

CO2 :: validate the functionality of various digital & analogue ICs

CO3 :: use basic measuring instruments and component specific ratings related to Electronics Engineering

CO4 :: design applications and analyze combinational and sequential circuits

CO5 :: communicate and contribute effectively as an individual and as a member of a team

CO6 :: build applications with at least one sensor module by programming Microcontroller board

List of Practicals / Experiments:

The Electronics Expedition

- Familiarization of electronics components, trainer module, input sources and display devices related to electronic circuits

KVL Voltage Hunt

- Validation and analysis of Kirchhoff's Voltage Law (KVL)

Diode Voltage Quest

- Analyze V-I characteristics of PN junction diode

Universal Gate Adventure

- Implementation and validation of logic gates and their design using universal gates

Adder Chronicles

- Design half adder, full adder and validate the combinational logic

Flip-Flop Warriors: NAND Edition

- Design and validate JK flip-flop with NAND gates

List of Projects

- Design and implementation of 2-bit ALU performing arithmetic and logical operations with display
- Design and implementation of Electronics Frills using shift register or counters
- Design and implementation of 2-dimensional memory controller using decoders
- Design and implementation of digital clock with dual digit hour, minute and second display
- Design and implementation of encription-decription keys for error free communication

References:

1. DIGITAL LOGIC AND COMPUTER DESIGN by M. MORRIS MANO, PEARSON

2. INTERNET OF THINGS by RAJ KAMAL, MCGRAW HILL EDUCATION

3. DIGITAL DESIGN PRINCIPLES AND PRACTICES PEARSON by JOHN F. WAKERLY, PEARSON

4. DIGITAL ELECTRONICS, PRINCIPLES, DEVICES AND APPLICATIONS by ANIL K. MAINI, JOHN WILEY & SONS

5. FUNDAMENTALS OF ELECTRICAL ENGINEERING AND ELECTRONICS by B.L.THERAJA, S Chand Publishing